

Source citations

All sources checked 7 October 2026. Web access in the preparation environment was search-based; each URL below appeared in a search result or comes from supplied project material. "Slides" lists where each source is cited.

Teaching materials: Stanford Biodesign, Harvard, Cambridge

- Burns L, Denend L, Zenios S. The Role of Physicians in Device Innovation: Critical Success Factor or Conflict of Interest? Stanford GSB case OIT105; 2011. <https://www.gsb.stanford.edu/faculty-research/case-studies/role-physicians-device-innovation-critical-success-factor-or-conflict> — Slides: 2, 16
- Cambridge Judge Business School. Centre for Health Leadership and Enterprise. <https://www.jbs.cam.ac.uk/centres/health/> — Slides: 4, 19
- Chao KZ, Riskin DJ, Krummel TM. A patient-centered, ethical approach to medical device innovation. Virtual Mentor (now AMA J Ethics). 2010;12(2):91–95. <https://journalofethics.ama-assn.org/article/patient-centered-ethical-approach-medical-device-innovation/2010-02> — Slides: 2, 16
- Chaturvedi J, Logan A, Narayan G, Kuttappa S. A structured process for unmet clinical need analysis for medical device innovation in India: early experiences. BMJ Innovations. 2015;1(3):81. <https://innovations.bmj.com/content/1/3/81> — Slides: 4
- Clarkson PJ. What has engineering design to say about healthcare improvement? Design Science. 2018;4:e17. <https://www.repository.cam.ac.uk/handle/1810/287837> — Slides: appendix only
- Cohen IG, Gerke S, Kramer DB. Ethical and legal implications of remote monitoring of medical devices. Milbank Q. 2020;98(4):1257–1289. <https://www.milbank.org/quarterly/articles/ethical-and-legal-implications-of-remote-monitoring-of-medical-devices/> — Slides: appendix only
- Gerke S, Babic B, Evgeniou T, Cohen IG. The need for a system view to regulate artificial intelligence/machine learning-based software as medical device. npj Digit Med. 2020. <https://hls.harvard.edu/bibliography/the-need-for-a-system-view-to-regulate-artificial-intelligence-machine-learning-based-software-as-medical-device> — Slides: 11, 21
- Harvard Catalyst. Regulatory Foundations, Ethics, and Law Program. <https://catalyst.harvard.edu/regulatory/> — Slides: appendix only
- Harvard Medical School Center for Bioethics. Fellowship in Bioethics: curriculum. <https://bioethics.hms.harvard.edu/education/fellowship-bioethics/curriculum> — Slides: 4
- Harvard Medical School. Policy on Conflicts of Interest and Commitment (2010 revision). <https://hms.harvard.edu/news/changes-conflicts-policy-heighten-transparency> — Slides: 16
- Kramer DB, Tan YT, Sato C, Kesselheim AS. Postmarket surveillance of medical devices: a comparison of strategies in the US, EU, Japan, and China. PLoS Med. 2013. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3815401/> — Slides: 22
- MRCT Center of Brigham and Women's Hospital and Harvard. Achieving Diversity, Inclusion, and Equity in Clinical Research: Guidance Document and Toolkit; 2020. <https://mrctcenter.org/diversity-in-clinical-research/> — Slides: 4, 14, 25
- MRCT Center Post-trial Continued Access Task Force. Post-trial, Continued Access Responsibilities to Investigational Significant-Risk Devices: Framework; 2025. https://mrctcenter.org/wp-content/uploads/2025/04/2025-04-23c-Responsibilities-Framework_Devices.pdf — Slides: 10
- MRCT Center. Accessibility by Design in Clinical Research Toolkit; 2023. <https://mrctcenter.org/resource/accessibility-by-design-abd-toolkit-release/> — Slides: 18
- MRCT Center. Clinical Research Glossary (plain-language definitions; v3.0, 2025). <https://mrctcenter.org/glossary/> — Slides: 10, 15
- MRCT Center. Return of Aggregate Results to Participants: Guidance Document and Toolkit, v3.1; 2017. <https://mrctcenter.org/wp-content/uploads/2023/03/2017-12-07-MRCT-Return-of-Aggregate-Results-Toolkit-3.1>

[pdf](#) — Slides: 16

- Petrie-Flom Center, Harvard Law School. Diagnosing in the Home: the ethical, legal and regulatory challenges of digital home health (2021–2024). <https://petrieflom.law.harvard.edu/diagnosing-in-the-home/> — Slides: 4
- Rangan VK. The Aravind Eye Hospital, Madurai, India: In Service for Sight. Harvard Business School case 593-098; 1993. — Slides: 19
- Royal Academy of Engineering, Academy of Medical Sciences, Royal College of Physicians. Engineering Better Care: a systems approach to health and care design and continuous improvement; 2017 (working group chaired by Prof P. John Clarkson, University of Cambridge). <https://raeng.org.uk/publications/reports/engineering-better-care> — Slides: 4, 6, 11
- Simon DA, Shachar C, Cohen IG. Unsettled liability issues for 'prediagnostic' wearables and health-related products. JAMA. 2022;328:1391. <https://hls.harvard.edu/bibliography/unsettled-liability-issues-for-prediagnostic-wearables-and-health-related-products> — Slides: 7
- Stanford Biodesign. Health Equity programme and resources. <https://biodesign.stanford.edu/programs/health-equity.html> — Slides: 14
- Stanford Biodesign. Justice, Equity, Diversity and Inclusion (JEDI). Student Guide to Biodesign; 2022. <https://biodesignguide.stanford.edu/wp-content/uploads/2022/08/JEDI-v1.pdf> — Slides: appendix only
- Stanford Biodesign. Need Statements. Student Guide to Biodesign; 2022. <https://biodesignguide.stanford.edu/wp-content/uploads/2022/07/Need-Statements-v2.pdf> — Slides: 7, 25
- Stanford Biodesign. Principled Decision-Making. Student Guide to Biodesign; 2022. <https://biodesignguide.stanford.edu/wp-content/uploads/2022/08/Principled-Decision-Making-v1.pdf> — Slides: 4, 20, 24
- Stanford Biodesign. Stanford Biodesign in India (Stanford-India Biodesign, 2007–2014; now School of International Biodesign, AIIMS and IIT Delhi). <https://biodesign.stanford.edu/content/dam/sm/biodesign/documents/programs/global-initiatives/founders-forum/Stanford-Bioesign-in-India.pdf> — Slides: 4
- Stanford Mussallem Center for Biodesign. Our Process: Identify, Invent, Implement. <https://biodesign.stanford.edu/about-us/process.html> — Slides: 4, 6
- University of Cambridge Centre for Engineering Better Care. Improving Improvement: a toolkit for Engineering Better Care. <https://cebc.eng.cam.ac.uk/engineering-better-care> — Slides: appendix only
- University of Cambridge Department of Engineering. Part IIB module 4G9: Biomedical Engineering / Biomedical Innovation (syllabus). <https://teaching.eng.cam.ac.uk/content/engineering-tripos-part-iib-4g9-biomedical-engineering-2024-25> — Slides: 18
- University of Cambridge Engineering Design Centre. Inclusive Design Toolkit and Exclusion Calculator (Clarkson, Waller, Coleman). <https://www-edc.eng.cam.ac.uk/research/inclusive-design> — Slides: 4, 18, 25
- University of Cambridge Innovation and Entrepreneurship. How to Guide 3: Spinouts; 2024. https://ie.cam.ac.uk/wp-content/uploads/2024/04/H2G-3-Spinouts_V2_ISSUU.pdf — Slides: 16
- University of Cambridge. Conflict of Interest Policy. https://www.governanceandcompliance.admin.cam.ac.uk/files/conflict_of_interest_policy.pdf — Slides: 2
- University of Cambridge. New Cambridge institute will support high-value medical device translation; January 2026. <https://www.cam.ac.uk/research/news/new-cambridge-institute-will-support-high-value-medical-device-translation> — Slides: 4, 9
- Yock PG, Zenios S, Makower J, Brinton TJ, Kumar UN, Watkins FTJ, et al. Biodesign: The Process of Innovating Medical Technologies. 2nd ed. Cambridge University Press; 2015 (Stanford Biodesign content). https://assets.cambridge.org/97811070/87354/frontmatter/9781107087354_frontmatter.pdf — Slides: appendix only

India: regulation, ethics, pricing and safety

- Advertising Standards Council of India. Consumer Complaints Council recommendations press release (June 2022): health advertisements assessed under ASCI Code Chapter I.1 (truthful, substantiated claims) and III.4. https://www.ascionline.in/wp-content/uploads/2022/09/asci_june-17_ccc_recomm_press_release.pdf — Slides: 20

- Business Today. Govt to ask Johnson & Johnson to compensate for faulty hip implants (Agarwal committee report, 19 Feb 2018: about 4,700 ASR surgeries in India 2004–2010; 1,080 patients traced); 25 August 2018. <https://www.businesstoday.in/amp/latest/economy-politics/story/johnson-johnson-faulty-hip-implants-surgeries-compensation-109085-2018-08-25> — Slides: 22
- CDSCO. Draft Guidance Document on Medical Device Software; 21 October 2025. https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadPublic_NoticesFiles/Draft%20guidance%20document%20on%20Medical%20Device%20Software%2021%2010%202025.pdf — Slides: 21
- CDSCO. Draft Guidance on Post Market Surveillance of Medical Devices; 2018. https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadPublic_NoticesFiles/2Draft%20Guidance%20on%20PMS.pdf — Slides: 17
- CDSCO. Form MD-12: application for licence to manufacture medical devices for clinical investigation, test, evaluation, examination, demonstration or training (licence in Form MD-13). https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/medical-device/12MD.pdf — Slides: 3, 10, 17, 23
- CDSCO. Form MD-22: application for permission to conduct clinical investigation (permission in Form MD-23). https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/medical-device/22MD.pdf — Slides: 3, 10, 17, 23
- CDSCO. Guidance Document on Medical Device Software under the Medical Devices Rules, 2017 (CDSCO/MD/GD/MDSW/01/2026); 21 July 2026. https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/Guidance-document-on-Medical-Device-Software-under-MDR-2017.pdf — Slides: 7, 17, 21, 25
- CDSCO. Regulatory pathway under the Medical Devices Rules, 2017 (overview). https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/medical-device/RegulatoryMDR-2017.pdf — Slides: 17
- Department of Pharmaceuticals. Uniform Code for Marketing Practices in Medical Devices, 2024 (circular of 6 September 2024; as amended on 30 April 2026). Voluntary code administered through industry associations. [https://pharma-dept.gov.in/sites/default/files/UCMDMD%20circular%201-2026+UCMDMP%20code%20as%20amended%20on%2030.4.2026%20\(1\).pdf](https://pharma-dept.gov.in/sites/default/files/UCMDMD%20circular%201-2026+UCMDMP%20code%20as%20amended%20on%2030.4.2026%20(1).pdf) — Slides: 16, 20
- Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 (Schedule includes 'rheumatism'). <https://www.indiacode.nic.in/indiacode/bitstream/123456789/1412/1/A1954-21.pdf> — Slides: 20
- India State-Level Disease Burden Initiative Neurological Disorders Collaborators. The burden of neurological disorders across the states of India: GBD 1990–2019. Lancet Glob Health. 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8295043/> — Slides: 5
- Indian Council of Medical Research. National Ethical Guidelines for Biomedical and Health Research Involving Human Participants. New Delhi: ICMR; 2017 (Section 5 informed consent; Section 6 vulnerability; Section 7.7 device trials). https://ethics.ncdirindia.org/asset/pdf/ICMR_National_Ethical_Guidelines.pdf — Slides: 6, 10, 15, 16, 25
- Indian Pharmacopoeia Commission (National Coordination Centre). Materiovigilance Programme of India (MvPI), launched 6 July 2015. <https://nhsrcindia.org/hi/node/10346> — Slides: appendix only
- Indian Pharmacopoeia Commission. Materiovigilance Programme of India (MvPI): About us. Launched 6 July 2015; helpline 1800-180-3024; mvpi-ipc@gov.in. <https://ipc.gov.in/mandates/materiovigilance-programme-of-india-mvpi/about-us.html> — Slides: 22, 25
- Indian Pharmacopoeia Commission. Medical Device Adverse Event Reporting Form, version 1.2 (open to manufacturers, importers, distributors, healthcare professionals and patients). [https://nhsrcindia.org/sites/default/files/Medical-Devices-Adverse-Event-Reporting-Form-\(Version-1.2\).pdf](https://nhsrcindia.org/sites/default/files/Medical-Devices-Adverse-Event-Reporting-Form-(Version-1.2).pdf) — Slides: 22
- Medical Council of India. Indian Medical Council (Professional Conduct, Etiquette and Ethics) Regulations, 2002, clause 6.8 (inserted 2009); operative after the NMC 2023 regulations were held in abeyance on 23 August 2023. <https://www.delhimedicalcouncil.org/pdf/EthicsRegulations-2002.pdf> — Slides: 16
- Medical Dialogues. NPPA continues price cap on orthopaedic knee replacement implants for another year, till 15 November 2026. <https://medicaldialogues.in/news/industry/medical-devices/nppa-continues-price-cap-on-orthopedic-knee-replacement-implants-for-another-year-till-november-15-2026-158881> — Slides: 19

- Ministry of Electronics and Information Technology. Digital Personal Data Protection Rules, 2025 (notified November 2025; phased commencement). <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc20251117695301.pdf> — Slides: 10
- Ministry of Health and Family Welfare. Medical Devices Rules, 2017 (as amended), consolidated text, CDSCO. [https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2022/m_device/mdr,%202017%20\(1\).pdf](https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2022/m_device/mdr,%202017%20(1).pdf) — Slides: 3, 7, 10, 17, 22, 25
- Ministry of Health and Family Welfare. New Drugs and Clinical Trials Rules, 2019 (ethics committee registration: Rules 7–8 with CDSCO; Rules 16–17 with DHR). https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2022/new_DC_rules/NEW%20DRUGS%20ANDctrS%20RULE,%202019.pdf — Slides: 3, 10, 17
- Pal CP, Singh P, Chaturvedi S, Pruthi KK, Vij A. Epidemiology of knee osteoarthritis in India and related factors. Indian J Orthop. 2016;50(5):518–522 (28.7% in a five-site community survey). <https://pmc.ncbi.nlm.nih.gov/articles/PMC5017174> — Slides: 5
- Press Information Bureau. Government caps trade margin up to 70% on price-to-distributor for pulse oximeters, BP monitors, nebulisers, digital thermometers and glucometers; 13 July 2021 (91% of 684 brands cut MRP, by up to 88%). <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1735430> — Slides: 19
- Press Information Bureau. Government fixes ceiling prices of knee implants (cobalt-chromium primary knee: average MRP Rs 1,58,324 to Rs 54,720, 65%; special metals 69%); 16 August 2017. <https://www.pib.gov.in/newsite/PrintRelease.aspx?relid=170040> — Slides: 19
- Press Information Bureau. Health Ministry approves compensation formula for hip implant cases (Central Expert Committee; base amount Rs 20 lakh adjusted for disability and age); 29 November 2018. <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1554266> — Slides: 22
- Press Information Bureau. NPPA ceiling prices for coronary stents: Rs 7,260 (bare-metal) and Rs 29,600 (drug-eluting); 13 February 2017. <https://www.pib.gov.in/newsite/PrintRelease.aspx?relid=158452> — Slides: 19
- TCTMD. Stents as essential medicine: India's cap on stent prices could have ripples around the globe; 2017 (withdrawal applications refused). <https://www.tctmd.com/news/stents-essential-medicine-indias-cap-stent-prices-could-have-ripples-around-globe> — Slides: 19
- The Wire. NMC puts new guidelines on doctors' ethics in abeyance; August 2023. <https://thewire.in/health/nmc-puts-new-guidelines-on-doctors-ethics-in-abeyance-after-imas-meeting-with-health-minister> — Slides: 16

Ethics frameworks, evidence and international comparison

- Appelbaum PS, Roth LH, Lidz C. The therapeutic misconception: informed consent in psychiatric research. Int J Law Psychiatry. 1982;5(3–4):319–329. <https://psychnews.psychiatryonline.org/doi/10.1176/appi.pn.2021.7.30> — Slides: 15
- Appelbaum PS, Roth LH, Lidz CW, Benson P, Winslade W. False hopes and best data: consent to research and the therapeutic misconception. Hastings Cent Rep. 1987;17(2):20–24. <https://philpapers.org/rec/APPFHA> — Slides: 15
- Bächlin M, Plotnik M, Roggen D, et al. Wearable assistant for Parkinson's disease patients with the freezing of gait symptom. IEEE Trans Inf Technol Biomed. 2010 (10 patients in the laboratory; 8 froze; 237 video-labelled events). <https://folia.unifr.ch/global/documents/222920> — Slides: 9, 27
- Bannuru RR, Osani MC, Vaysbrot EE, et al. OARSI guidelines for the non-surgical management of knee, hip, and polyarticular osteoarthritis. Osteoarthritis Cartilage. 2019 (electromagnetic therapy strongly recommended against). https://www.esceo.org/sites/esceo/files/pdf/Bannuru_O%26C_OARSIguidelines_2019.pdf — Slides: 13, 20, 27
- Centers for Medicare & Medicaid Services. Open Payments: public reporting of industry payments to physicians and teaching hospitals. <https://www.cms.gov/files/document/physician-fact-sheetpdf> — Slides: 16
- Chen L, Duan X, Xing F, et al. Effects of pulsed electromagnetic field therapy on pain, stiffness and physical function in patients with knee osteoarthritis: a systematic review and meta-analysis. J Rehabil Med.

- 2019;51(11):821–827 (no pain advantage; function improved). <https://medicaljournals.se/jrm/content/abstract/10.2340/16501977-2613> — Slides: 13, 27
- CIOMS with WHO. International Ethical Guidelines for Health-related Research Involving Humans. 4th ed. Geneva; 2016 (Guideline 9 consent; Guideline 25 conflicts of interest). <https://cioms.ch/wp-content/uploads/2017/01/WEB-CIOMS-EthicalGuidelines.pdf> — Slides: 15
 - European Commission. Proposal for a targeted revision of the medical device regulations, COM(2025) 1023 final, 16 December 2025 (under negotiation). <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52025PC1023> — Slides: 17
 - General Medical Council (UK). Making and using visual and audio recordings of patients (effective 2011; updated December 2024). International comparator. <https://www.gmc-uk.org/professional-standards/the-professional-standards/making-and-using-visual-and-audio-recordings-of-patients> — Slides: 15
 - Ginis P, Nackaerts E, Nieuwboer A, Heremans E. Cueing for people with Parkinson's disease with freezing of gait: a narrative review. *Ann Phys Rehabil Med*. 2018;61(6):407–413. <https://pubmed.ncbi.nlm.nih.gov/28890341/> — Slides: 27
 - IEC 62366-1:2015+AMD1:2020. Medical devices — Part 1: Application of usability engineering to medical devices. <https://webstore.iec.ch/en/publication/67220> — Slides: 11, 18
 - Independent Medicines and Medical Devices Safety Review (chair Baroness Cumberlege). First Do No Harm; 8 July 2020. https://www.immdsreview.org.uk/downloads/IMMDSReview_Web.pdf — Slides: 16, 22
 - Institute of Medicine; Lo B, Field MJ, eds. Conflict of Interest in Medical Research, Education, and Practice. National Academies Press; 2009. <https://www.nationalacademies.org/read/12598/chapter/11> — Slides: 2
 - ISO 14155:2026. Clinical investigation of medical devices for human subjects — Good clinical practice (supersedes ISO 14155:2020). <https://www.iso.org/standard/83968.html> — Slides: 17
 - ISO 14971:2019. Medical devices — Application of risk management to medical devices. <https://www.iso.org/standard/72704.html> — Slides: 11, 25
 - Johnson J, Rogers W. Innovative surgery: the ethical challenges. *J Med Ethics*. 2012;38(1):9–12. <https://jme.bmj.com/content/38/1/9> — Slides: 2
 - Kadakia KT, et al. Class I recalls of cardiovascular devices between 2013 and 2022: a cross-sectional analysis. *Ann Intern Med*. 2024;177(11) (19.1% had premarket clinical testing). <https://www.acpjournals.org/doi/10.7326/ANNALS-24-00724> — Slides: 17
 - Kellmeyer P, Cochrane T, Müller O, et al. The effects of closed-loop medical devices on the autonomy and accountability of persons and systems. *Camb Q Healthc Ethics*. 2016;25(4):623–633. <https://www.research-collection.ethz.ch/handle/20.500.11850/126117> — Slides: 11, 21
 - Li S, Yu B, Zhou D, et al. Electromagnetic fields for treating osteoarthritis. *Cochrane Database Syst Rev*. 2013; (12):CD003523 (9 trials, 636 adults: pain probably improves about 15/100 more than sham; function uncertain). https://www.cochrane.org/evidence/CD003523_electromagnetic-fields-treatment-osteoarthritis — Slides: 13, 27
 - Lundh A, Lexchin J, Mintzes B, Schroll JB, Bero L. Industry sponsorship and research outcome. *Cochrane Database Syst Rev*. 2017;(2):MR000033 (drug and device studies: favourable conclusions RR 1.34, 95% CI 1.19–1.51). <https://pubmed.ncbi.nlm.nih.gov/28207928/> — Slides: 13, 16
 - Markovic L, Wagner B, Crevenna R. Effects of pulsed electromagnetic field therapy on outcomes associated with osteoarthritis: a systematic review of systematic reviews. *Wien Klin Wochenschr*. 2022 (heterogeneous schemes; no sufficient proof of efficacy). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9213303/> — Slides: 27
 - McCulloch P, Altman DG, Campbell WB, et al. No surgical innovation without evaluation: the IDEAL recommendations. *Lancet*. 2009;374:1105–1112 (IDEAL Collaboration; 2019 update adds Pre-IDEAL). <https://www.ideal-collaboration.net/the-ideal-framework/> — Slides: 9
 - NICE. Osteoarthritis in over 16s: diagnosis and management (NG226); October 2022 (do not offer listed electrotherapies: insufficient evidence of benefit). <https://www.nice.org.uk/guidance/ng226/chapter/Recommendations> — Slides: 27

- Nieuwboer A, et al. Cueing training in the home improves gait-related mobility in Parkinson's disease: the RESCUE trial. *J Neurol Neurosurg Psychiatry*. 2007;78(2):134 (n=153; small gait gains and lower freezing severity in freezers; fall counts not measured). <https://jnnp.bmj.com/content/78/2/134> — Slides: 27
- Pardoel S, Kofman J, Nantel J, Lemaire ED. Wearable-sensor-based detection and prediction of freezing of gait in Parkinson's disease: a review. *Sensors*. 2019;19(23):5141. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6928783/> — Slides: 9, 27
- Regulation (EU) 2017/745 on medical devices, Art. 61 and Annex XIV Part B (post-market clinical follow-up); transition extended to 2027/2028 by Regulation (EU) 2023/607. <https://eur-lex.europa.eu/eli/reg/2017/745/2023-03-20> — Slides: 17
- Sedrakyan A, Campbell B, Merino JG, Kuntz R, Hirst A, McCulloch P. IDEAL-D: a rational framework for evaluating and regulating the use of medical devices. *BMJ*. 2016;353:i2372. <https://pubmed.ncbi.nlm.nih.gov/27283585/> — Slides: 9, 25, 26
- Silva de Lima AL, Evers LJW, Hahn T, et al. Freezing of gait and fall detection in Parkinson's disease using wearable sensors: a systematic review. *J Neurol*. 2017 (sensitivity 73–100%, specificity 67–100%). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5533840/> — Slides: 9, 27
- Sjoding MW, Dickson RP, Iwashyna TJ, Gay SE, Valley TS. Racial bias in pulse oximetry measurement. *N Engl J Med*. 2020;383(25):2477–2478. doi:10.1056/NEJMc2029240. <https://www.nejm.org/doi/full/10.1056/NEJMc2029240> — Slides: 14
- Tubach F, Ravaud P, Baron G, et al. Evaluation of clinically relevant changes in patient reported outcomes in knee and hip osteoarthritis: the minimal clinically important improvement. *Ann Rheum Dis*. 2005;64(1):29–33 (knee OA pain MCII –19.9 mm, –40.8%). <https://ard.bmj.com/content/64/1/29.full.pdf> — Slides: 13, 26, 27
- US FDA. Information Sheet: Significant Risk and Nonsignificant Risk Medical Device Studies (21 CFR 812). <https://www.fda.gov/files/about%20fda/published/Significant-Risk-and-Nonsignificant-Risk-Medical-Device-Studies---Information-Sheet.pdf> — Slides: 17
- US FDA. Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence-Enabled Device Software Functions. Final guidance, December 2024 (revised version reported August 2025). <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/marketing-submission-recommendations-predetermined-change-control-plan-artificial-intelligence> — Slides: 21
- US FDA. Pulse Oximeters for Medical Purposes: Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations. Draft guidance, 7 January 2025 (diverse skin-tone testing, Monk Skin Tone scale). <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/pulse-oximeters-medical-purposes-non-clinical-and-clinical-performance-testing-labeling-and> — Slides: 14
- US HHS. Protection of Human Subjects, 45 CFR 46.116(a)(5)(i): consent must begin with a concise presentation of key information. International comparator. <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-46/subpart-A/section-46.116> — Slides: 15
- Wexler A, Largent E. Ethical considerations for researchers developing and testing minimal-risk devices. *Nat Commun*. 2023;14. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10122640/> — Slides: 10
- World Medical Association. Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Participants. 75th General Assembly, Helsinki, October 2024 (para 27: dependent relationship; para 34: post-trial provisions). <https://www.wma.net/policies-post/wma-declaration-of-helsinki/> — Slides: 10, 15, 25
- Yang X, He H, Ye W, Perry TA, He C. Effects of pulsed electromagnetic field therapy on pain, stiffness, physical function, and quality of life in patients with osteoarthritis: a systematic review and meta-analysis. *Phys Ther*. 2020;100(7):1118–1131. <https://pubmed.ncbi.nlm.nih.gov/32251502/> — Slides: 13, 27

Case material and media (supplied)

- Ahilaya Biomedicals Pvt Ltd. FoGO BIRAC BIG project presentation (supplied, 2026): alpha prototype (3 iterations, Jan 2024–Jun 2025); subject-wise F1 0.83–0.85 and cross-dataset F1 0.74–0.80 on Daphnet, CuPiD and Turning-in-place; 0.15–0.18 false alarms per minute; vibrotactile bench study; IEC 60601-1-2 EMC pre-

compliance (Jul 2026); CDSCO test licence Form MD-13 for 25 units (10 Aug 2026); clinical proof-of-concept planned at AIIMS Bhubaneswar; 12 s demonstration clip (volunteer, simulated freeze, face obscured). Project-reported; not peer-reviewed. — *Slides: 1, 8, 9, 11, 17, 18, 19, 21, 23, 26, 27*

- Swa knee_Leaflet.pdf. Supplied product material: 45-minute daily sessions over 45 days (illustration of time burden, not a prescription). — *Slides: 12*
- Swayogya Rehab Solutions. SwaKnee evidence page: company-reported comparative 45-day study, device n=40 vs comparison n=42, ≈32% vs ≈14% average VAS pain reduction from baseline. Rounded figures as transcribed in the supplied original deck; re-check the live page before presenting. <https://www.swayogya.in/evidence.html> — *Slides: 13, 20, 26*
- Swayogya Rehab Solutions. SwaKnee explainer video on YouTube (company material; optional, longer than the embedded clip). <https://www.youtube.com/watch?v=IfQyVm8Kftc> — *Slides: 12*
- Swayogya website assets (ivikajha/swayogya_website): product photographs and three 'How to use' step videos, edited into a 20 s silent clip with step captions; no faces shown. <https://www.swayogya.in/product.html> — *Slides: 1, 12*

Notes on attribution

- *Biodesign: The Process of Innovating Medical Technologies* is Stanford Biodesign content published by Cambridge University Press; it is cited as Stanford material.
- *Engineering Better Care* (2017) was published by the Royal Academy of Engineering, Academy of Medical Sciences and Royal College of Physicians; its working group was chaired by Prof P. John Clarkson (University of Cambridge), and Cambridge's Centre for Engineering Better Care maintains the toolkit.
- The IDEAL and IDEAL-D frameworks are Oxford-led (IDEAL Collaboration), not Cambridge.
- FoGO results are project-reported (Ahilaya Biomedicals); SwaKnee results are company-reported (Swayogya); neither has been independently verified or peer-reviewed. No cited institution has evaluated or endorsed either device.